

Building AI Apps with Gemini and Google AI Studio

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What we'll cover

AGENDA

01 — Introduction to Gemini

The model family and what changed with Gemini 3

02 — Google AI Studio Workflow

Playground → Build → SDK → Vertex

03 — Five Lessons from Gemini 3

Multimodality, long context, grounding, generative UI, context engineering

04 — Building & Agentic Workflows

Demos, the SDK, and where this is going

01

Introduction to Gemini

The model family, the context, and what actually changed with Gemini 3

DEMO — Whiteboard: What is Gemini?

Google's frontier multimodal model family

What is Gemini

WHAT IS GEMINI

- **Native multimodality** — One model, one call: text + image + video + audio
- **Long context** — 1M+ tokens — used as working memory, not storage
- **Tool-using by default** — Search grounding, code execution, URL context, function calling
- **Family of sizes** — Pro · Flash · Flash-Lite — pick by cost/latency/quality

Which model to use

MODEL PICKER

Gemini 3.5 Flash

DEFAULT

GA at I/O 2026. Thinking model, runs on the free tier — your workhorse.

Gemini 3.1 Flash-Lite

BUDGET

Cheapest and fastest — high-volume, latency-sensitive work.

Gemini 3.1 Flash Image

IMAGES

Nano Banana — fast image generation & editing (gemini-3.1-flash-image).

Gemini 3.1 Pro

PREVIEW · PAID

Strongest reasoning — but zero free quota, billing required.

AI work is now input design work.

COURSE THESIS

Multimodal inputs · curated context · specified output format.

Effort moves away from prompt-craft toward **designing what the model sees and emits.**

02

The AI Studio Workflow

How real builders actually go from idea to deployed prototype

AI Studio at a glance

AISTUDIO.GOOGLE.COM

1. **Playground** — System instructions, model picker, parameters, tool toggles.
Where you start.
2. **Build** — Vibe coding: scaffold a React + Node app from a prompt. Annotation Mode FTW.
3. **Stream Realtime** — Live API: voice / webcam / screen-share with low-latency bidirectional audio.
4. **Generate Media** — Imagen, Veo, Lyria RealTime, native speech.
5. **Library** — Saved prompts and projects. ⚠️ no version control — export to Git.

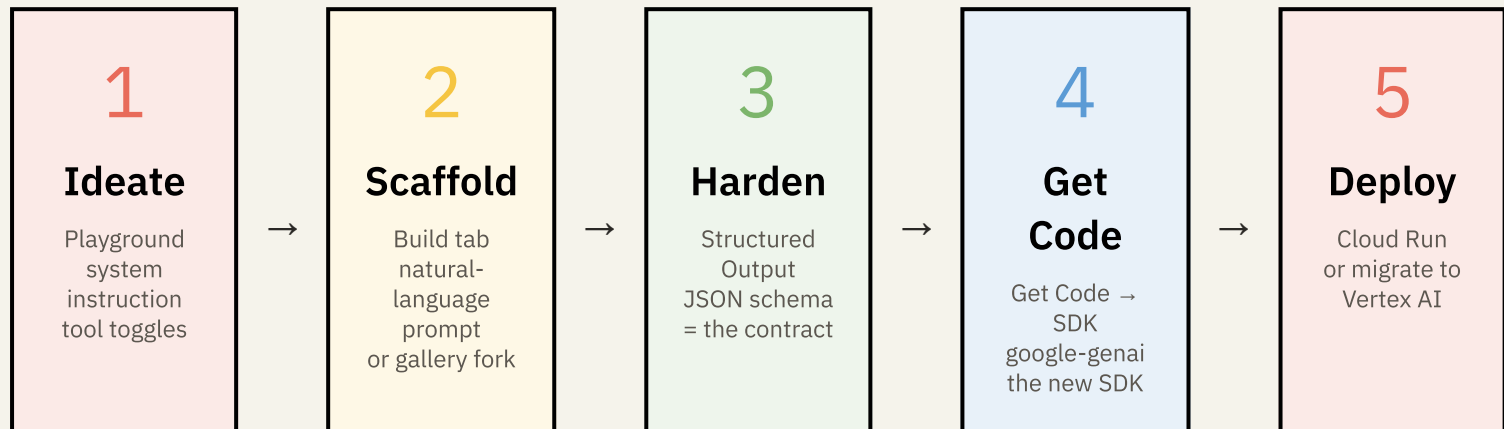
Note: the Tune section is effectively defunct — current Gemini models aren't fine-tunable in the UI.

DEMO — AI Studio Walkthrough

Playground · Build · Stream Realtime · Generate Media · Library

From idea to deployed prototype

THE CANONICAL WORKFLOW



Patterns power users actually use

PLAYGROUND

System instruction

Carries role + explicit prohibitions ("do not speculate") — measurably reduces hallucinations

Structured Output toggle

JSON schema, not prose. Gemini quirk: fields are OPTIONAL by default — tag required ones

Thinking level

Sidebar dial most beginners miss. "high" = materially better SVG/complex reasoning

Tool toggles

Search · Code Execution · URL Context · Function Calling — checkboxes, not separate APIs

Multimodal drop-in

PDFs, videos, screenshots. No preprocessing. Long-audio: compress with ffmpeg if needed

Get Code

Two clicks → working Python/JS/REST snippet using google-genai

Logan Kilpatrick's prototyping tricks

POWER-USER MOVES

Clone-then-build Screenshot an existing UI · drop it in · type "clone this UI exactly" · iterate from the scaffold rather than from zero.

Single-prompt design exploration "Add a widget so I can click through these styles: [5 options]" Explore variants without spawning N sessions.

Connect Drive early AI Studio sessions don't persist by default — connect to Google Drive or you'll lose your work.

DEMO — Nano Banana Pro in the Playground

Diagrams, dashboards, infographics, brand collateral

"Vibe coding" — what it really is

BUILD MODE

What you get

- React + Node.js scaffold (Next.js / Angular as options)
- Three-panel IDE: agent reasoning · code · live preview
- Full npm; agent auto-installs deps & self-corrects type errors
- **Annotation Mode:** click on the preview, type the change

Deployment exits

- Download ZIP → take to local editor
- Push to GitHub → real version control
- Deploy to Cloud Run → one-click, secrets stay server-side, scale-to-zero pricing

Reality check: "pair programming with a fast senior dev." Prototyping accelerator, not production deployment.

DEMO — Build mode: fork & deploy

Annotation Mode → Cloud Run

What's a UI toggle in AI Studio

NATIVE TOOLS

- ✓ **Search Grounding** — cite sources, real-time facts
- ✓ **Code Execution** — sandboxed Python in the loop
- ✓ **URL Context** — reads URLs in your prompt
- ✓ **Function Calling** — tools you define

- ✓ **Files API** — PDFs, videos, audio
- ✓ **Live API** — voice/video/screen, WebSocket
- ✓ **Imagen / Veo / Lyria** — image, video, music
- ✓ **MCP** — Maps example bundled

Almost everything is a checkbox now — convenience vs. programmatic control, not access.

The SDK trap

CRITICAL SDK MIGRATION

▲ Half the tutorials online use `google-generativeai` — deprecated Nov 2025, broken on Live API, Veo, and Files API v2. Use only `google-genai`.

Use this — what AI Studio exports

```
from google import genai

client = genai.Client()
response = client.models.generate_content(
    model="gemini-3.5-flash",
    contents="Hi",
)
```

DEMO — Get Code → google-genai SDK

From AI Studio prompt to a Python notebook

The data-privacy switch

DATA & PRIVACY

AI Studio (free)

Trains on prompts: Yes

Reviewers see data: Possible

Compliance: —

Setup: Zero

AI Studio (Cloud Billing on)

Trains on prompts: No

Reviewers see data: No

Compliance: Limited

Setup: Low

Vertex AI

Trains on prompts: No (default)

Reviewers see data: No

Compliance: HIPAA / SOC2

Setup: High (GCP/IAM)

Enabling Cloud Billing on the API key flips training off — even within free quotas.

AI Studio → Vertex AI

WHEN TO GRADUATE

- Hit 429s on free key
- Org requires IAM auth
- Region / data residency
- MLOps tooling

- Need an SLA
- Regulated industry
- Plaintext API keys are out
- Model Registry, Evaluation, Monitoring

Good news: SDK migration is trivial — both sides use unified google-genai.
One-line endpoint swap.

What to watch out for

GOTCHAS

Hidden thinking tokens

Flash thinks by default (medium) — a short prompt can bill hundreds of invisible reasoning tokens. On Gemini 3.x you dial it with `thinking_level` (low/medium/high); there's no `thinking_budget=0` off-switch anymore — low minimizes it.

No prompt version control

AI Studio is a scratchpad — nothing is saved. Export to Git or notebooks immediately.

Pro tiers are paid-only

Free key + Pro model = 429 — `limit: 0`. Prototype on Flash; the strongest model has no free quota.

"Get Code" is a starting point

Exported code has no error handling, streaming, or auth. It's a scaffold, not production-ready.

Product fragmentation

AI Studio · Vertex AI Studio · Firebase Studio · Gemini CLI · Jules · Opal · Stitch · Colab AI · NotebookLM — pick one and stay focused.

03

Five Lessons from Gemini 3

What changed — and how each shift translates to API/AI Studio capability

What actually changed

THE FIVE LESSONS

01

Unified multimodality

One model, one call

02

Long context = working memory

Not a storage bin

03

Grounding

Turn scattered data into a knowledge base

04

Generative UI

Output format = the surface

05

Context > prompt engineering

Design what the model sees

+1

Reduced sycophancy

Ask for critique, not approval

Unified multimodality

LESSON 01

One model, one call. Video + audio + image processed jointly — not as a pipeline of single-modality tools.

Demo prompt

"I just recorded a walkthrough on toggling smart features in Gmail. Watch the recording and turn it into a clean step-by-step checklist I can hand to a new hire."

Use cases

- Screen recording → SOP checklist
- User-interview video → "list every pause > 3s and what was on screen"
- Dense PDF → infographic via Nano Banana Pro

DEMO — Multimodality: video → SOP

One model, one call. No frame extraction.

Long context = working memory

LESSON 02

The 1M+ window already existed. What changed is retrieval quality — ~60% better at deep needle-in-haystack reasoning.

Storage bin (wrong)

"Summarize these documents."

Generic summary. The model glances. Information sits in the window unused.

Working memory (right)

"Find the three biggest discrepancies between management strategy in the earnings call and what the financial PDFs actually show."

Cross-source reasoning. The model connects evidence buried in different files.

DEMO — Long context as working memory

Same uploads, two prompts: "summarize" vs "find contradictions"

Grounding

LESSON 03

Workspace, Search, URL Context, your own data — turn scattered digital history into a single queryable knowledge base.

Power combo: URL Context + Search grounding

Search finds relevant URLs · URL Context reads them · lightweight RAG substitute with zero infra.

(TDS: "another nail in RAG's coffin" for prototypes.)

Rule of thumb: if you'd normally spend > 10 minutes hunting through emails or docs to reconstruct context, ask Gemini first.

DEMO — Grounding with Search + URL Context

A lightweight RAG substitute, zero infra

Generative UI

LESSON 04

Screen Spot Pro: 11.4% → 72.7%. Output format becomes the artifact — calculators, dashboards, quizzes. The surface designs around the task.

Before

"Compare these 3 platforms."

→ static comparison table

After

"Compare these 3 platforms. Give me a dashboard with a slider for budget and a toggle for region."

→ interactive HTML tool with revenue calculator, break-even tab, recommendation quiz

Pro tip: explicitly request the controls you want — sliders, toggles, filters.

DEMO — Generative UI: an interactive HTML tool

Plain text vs. dashboard with sliders and toggles

Context engineering > prompt engineering

LESSON 05

Gemini can guess your tone, format, and length. It can't guess your facts. Spend effort on context, not phrasing.

Old style

"Act as a punchy thought-leadership writer. Voice: confident but not arrogant. Use 2-3 short paragraphs..."

Adjective soup. Generic results.

New style

"Here are 3 of my VP's previous LinkedIn posts. Based on these, rewrite this Q4 report as a LinkedIn post."

Ground-truth examples. Output sounds like the VP.

DEMO — Context engineering: VP LinkedIn voice

Three previous posts as ground-truth examples

Reduced sycophancy

BONUS LESSON

Google explicitly trained Gemini 3 to push back. Use it.

Red-team prompt

"Identify storytelling weaknesses and logical contradictions between the sections of this report. Predict the pushback I'd receive from leadership."

Ask for critique, not approval. Adopt a critical persona for sharper feedback.

DEMO — Red-team your own deck

Find storytelling weaknesses and contradictions

04

Building & Agentic Workflows

Demos, the SDK, and where this is going

Fork from the gallery

[AISTUDIO.GOOGLE.COM/APPS/BUNDLED](https://aistudio.google.com/apps/bundled)

Bundled apps are the intended starting point — preview, fork, modify, deploy.

GeoSeeker

Spatial understanding · location reasoning

Maps Travel Planner

Maps + Gemini → trip planning

Structured Data Extractor

Forms / docs → JSON in one click

Nano Banana Pro

Diagrams · dashboards · brand collateral ·
infographics

DEMO — Bundled apps tour

GeoSeeker · Maps Travel Planner · Structured Data Extractor

Visual communication just became machine-readable

NANO BANANA PRO

Diagrams & Layouts

Flowcharts · Animated diagrams · Dashboards · Editorial spreads · Blueprints · Structured one-pagers

Data Visualization

Visualize academic papers · Earnings reports → charts · Representation transforms

Marketing & Brand

Brand palettes & logos · Concept art · Brand collateral

Agentic Workflows

Agents generating diagrams · Visual PDF summaries · Updating onboarding assets

05

Agentic Workflows with Gemini

DEMO — Building agents with Gemini

Whiteboard + code

Design the inputs.

Specify the outputs.

Trust the model.

CLOSING THESIS

Resources & References

- github.com/EnkrateiaLucca/oreilly-live-trainings
- ai.google.dev
- aistudio.google.com